
✅ 1 Overview Section

🎬 Video title

Create an AWS S3 Bucket Using Terraform (it's simple) ([YouTube](#))

📌 Main objective of the video

- Demonstrate how to **provision an AWS S3 bucket** using Terraform.
- Walk through resource definition, init/plan/apply workflow, and life cycle properties. ([DEV Community](#))

📋 Prerequisites assumed

- Terraform installed and configured with AWS provider.
- AWS credentials configured (env vars or AWS CLI).
- Basic understanding of Terraform workflow (init/plan/apply). ([DEV Community](#))

🔧 Tools used

- Terraform CLI (init, plan, apply, destroy)
- AWS provider
- AWS S3 service

✅ 2 Core Concepts Explained

💠 AWS S3

💡 Definition

Amazon Simple Storage Service (S3) is an object storage service for storing and retrieving any amount of data. ([DEV Community](#))

⚙️ Why It's Used

- Durable and scalable file storage.
- Used for static website hosting, backups, logs, artifacts. ([DEV Community](#))

Real-world relevance

S3 is one of the most frequently provisioned resources in AWS automation workflows. ([DEV Community](#))

Terraform Resource Block

Definition

A block in Terraform config that declares an infrastructure resource to be created. ([DEV Community](#))

Why It's Used

To describe what cloud resource you want Terraform to manage. ([DEV Community](#))

AWS relation

Terraform uses the AWS provider to handle AWS resources like S3 buckets. ([DEV Community](#))

Lifecycle & Idempotency

Definition

Terraform's ability to notice if resource configuration changes and only apply differences. ([DEV Community](#))

Why It's Used

Ensures resources aren't recreated unnecessarily and operations are predictable. ([DEV Community](#))

Code Breakdown (VERY IMPORTANT)

Terraform config for an S3 bucket

```
provider "aws" {  
  region = "us-west-2"  
}
```

```
resource "aws_s3_bucket" "my_bucket" {
```

```
bucket = "my-unique-bucket-name-12345"
```

```
acl = "private"
```

```
tags = {
```

```
  Name      = "MyBucket"
```

```
  Environment = "Dev"
```

```
}
```

```
}
```

Explanation line-by-line

- **provider "aws" {}** — configures Terraform to use AWS APIs; region tells AWS where to create resources.
- **resource "aws_s3_bucket" "my_bucket" {}** — declares a new S3 bucket resource named my_bucket.
- **bucket = ...** — the unique bucket name. *S3 bucket names must be globally unique across all AWS users.* ([DEV Community](#))
- **acl = "private"** — sets the bucket to private (default best practice).
- **tags {...}** — optional human-friendly metadata on the bucket.

 ****Important »**

- S3 bucket names must be **globally unique** and all lowercase with no spaces. ([DEV Community](#))
- Terraform will create the bucket only if it doesn't already exist. ([DEV Community](#))

Commands Used

Command	Purpose
terraform init	Download provider plugins & initialize directory
terraform plan	Preview changes Terraform will make

Command	Purpose
terraform apply	Create/modify the AWS resources
terraform destroy	Remove created resources

Flags discussed

- -auto-approve — bypass manual y/n confirmation (used for quick lab runs). ([DEV Community](#))
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5 AWS Architecture Explained

Text-based architecture diagram derived from this exercise:

User (Terraform CLI) → Terraform Plan/Apply → AWS API → S3 Bucket

- Terraform sends API requests to AWS.
 - AWS provisions the S3 bucket.
 - Terraform state reflects the resource. ([DEV Community](#))
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6 Best Practices Mentioned

- Choose a **unique bucket name** to avoid provisioning failure. ([DEV Community](#))
 - Use clear **tags** (Name, Environment). ([DEV Community](#))
 - Don't hard-code sensitive info in .tf files. Auth via AWS CLI profiles or environment vars. ([DEV Community](#))
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7 Important Interview Questions From This Video

Conceptual

- What is an AWS S3 bucket?
- Why do S3 bucket names need to be globally unique?

Scenario-based

- What happens if the S3 bucket name is already taken during Terraform apply?

Debugging

- How to fix an InvalidBucketName error in Terraform? (likely due to naming rules)
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8 Common Errors / Debugging

Invalid bucket name

Cause: Bucket name not unique or malformed.

Fix: Change to a unique, valid name. ([DEV Community](#))

ACL misconfiguration

Cause: Incorrect ACL string

Fix: Use valid ACL values like private, public-read.

9 Summary for Quick Revision

- Terraform resource blocks define AWS resources to manage. ([DEV Community](#))
 - S3 bucket provisioning is basic but foundational. ([DEV Community](#))
 - Always run terraform plan before apply. ([DEV Community](#))
 - Unique names and correct ACLs are key for S3. ([DEV Community](#))
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10 Personal Notes Section

My Notes:

- First real AWS resource provisioning: an S3 bucket. ([YouTube](#))
 - S3 bucket requires a **globally unique name**, enforced by AWS. ([DEV Community](#))
 - Standard Terraform workflow used: init → plan → apply → destroy. ([DEV Community](#))
 - Demonstrated idempotency — running apply again with no changes results in “no changes”. ([DEV Community](#))
 - Tags help identify environment and resource purpose. ([DEV Community](#))
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